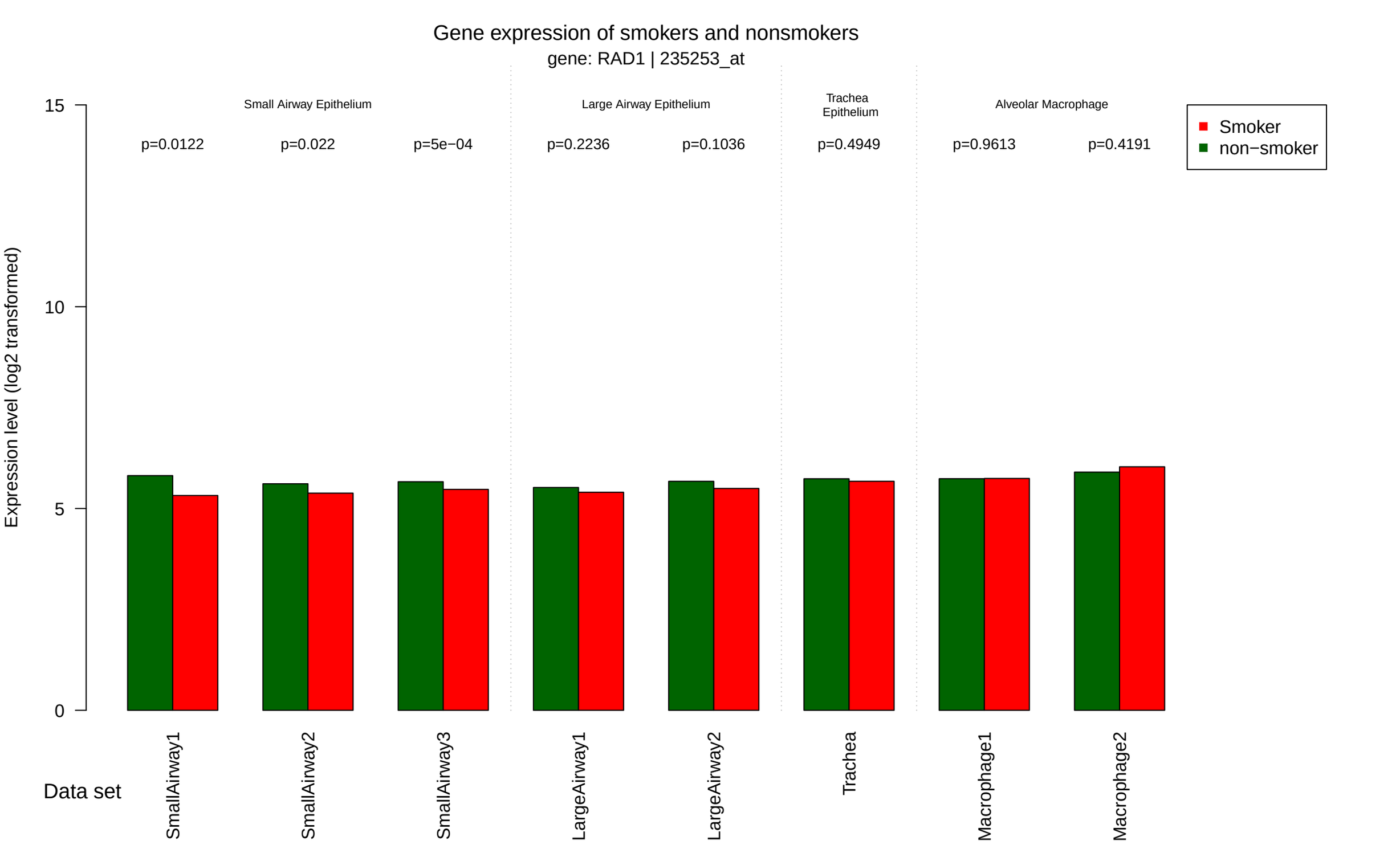
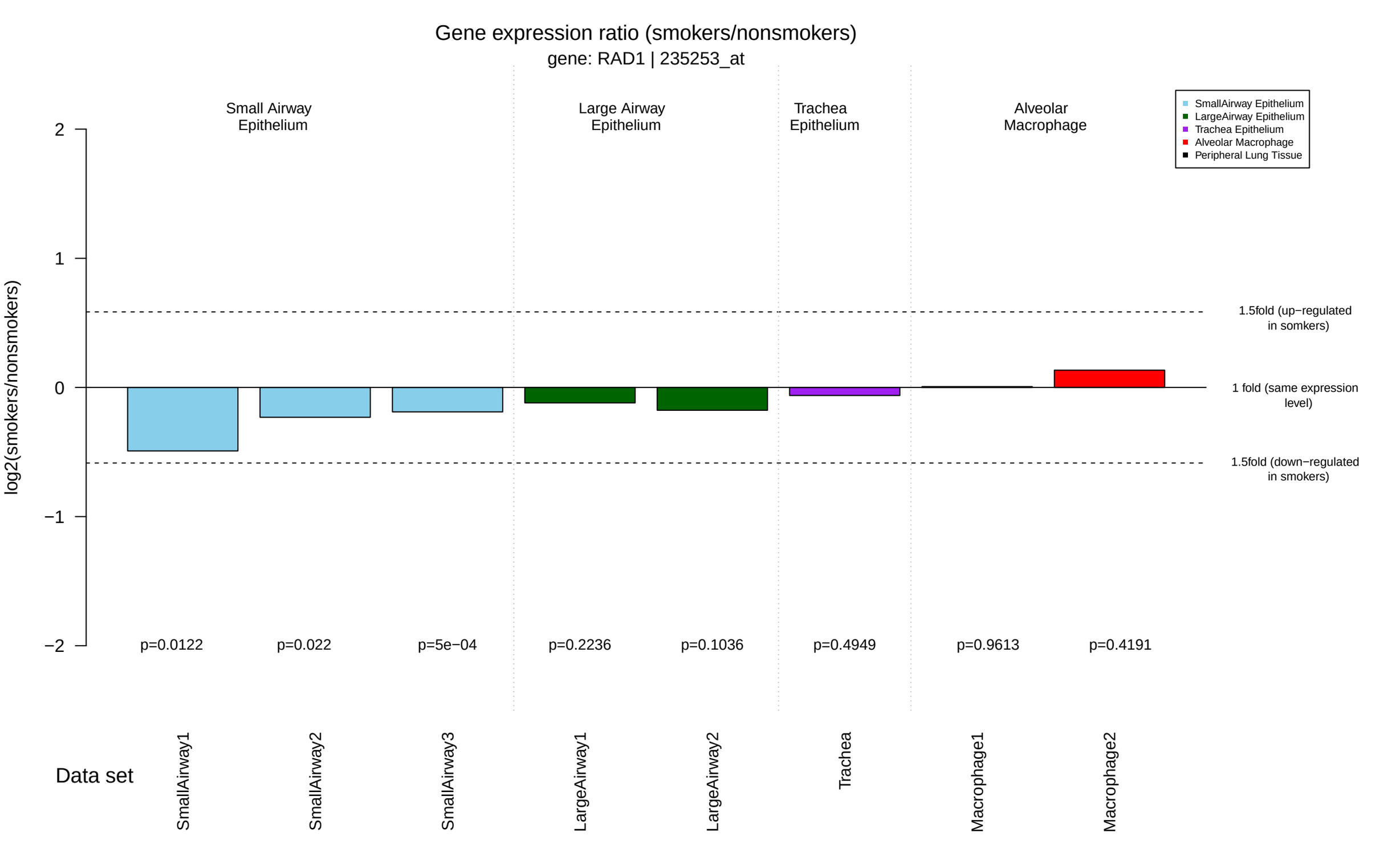
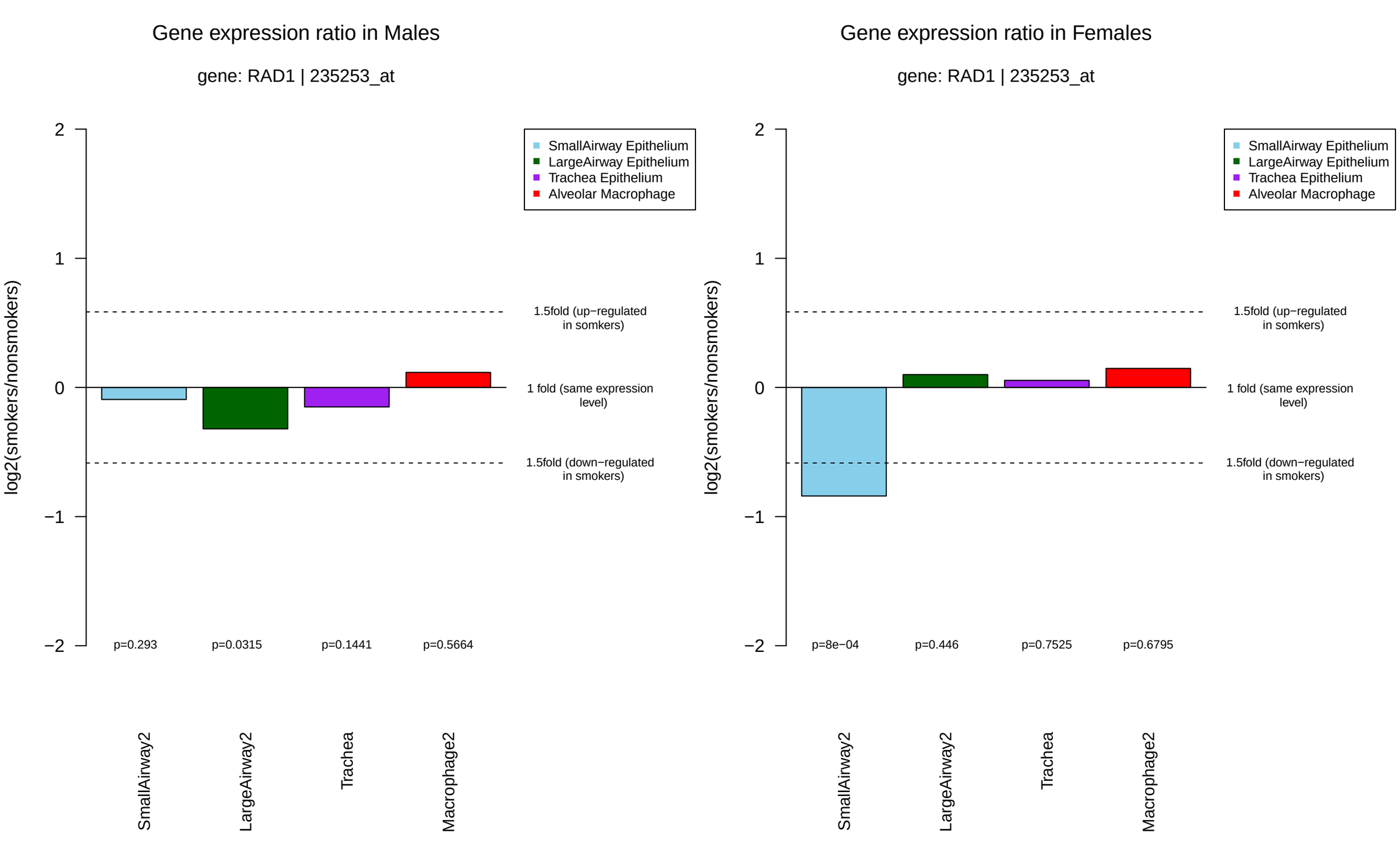
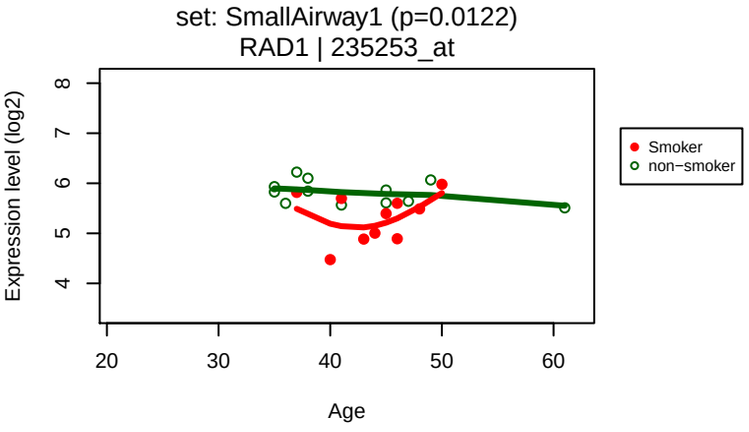
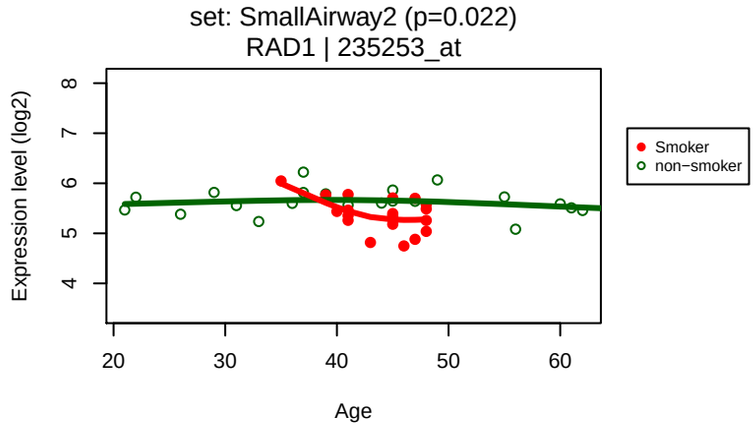
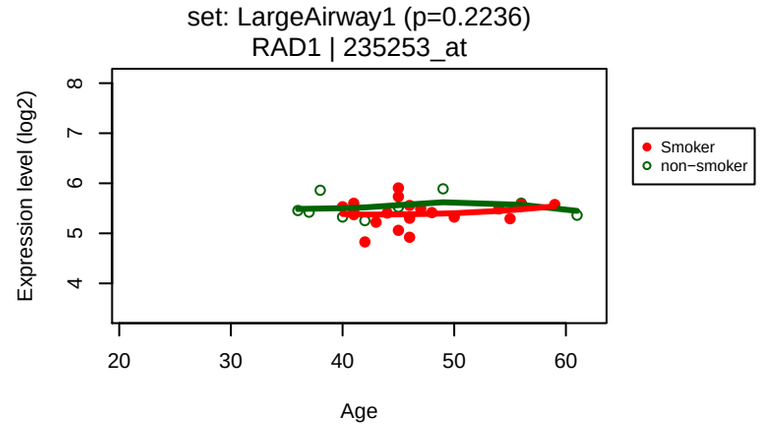

Gene expression of smokers and nonsmokers
gene: RAD1 | 235253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0122
p=0.022
p=5e−04
p=0.2236
p=0.1036
p=0.4949
p=0.9613
p=0.4191
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

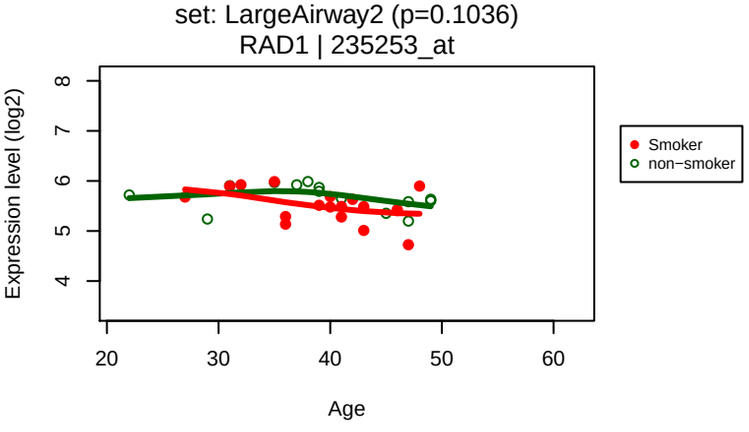

Gene expression ratio (smokers/nonsmokers)
gene: RAD1 | 235253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0122
p=0.022
p=5e−04
p=0.2236
p=0.1036
p=0.4949
p=0.9613
p=0.4191
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

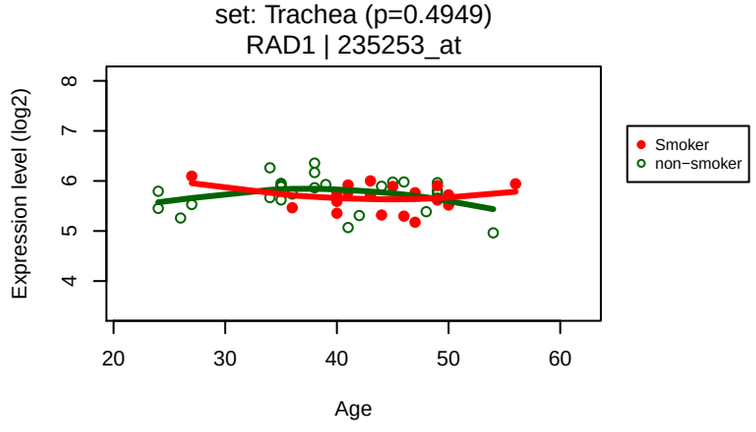

Gene expression ratio in Males
gene: RAD1 | 235253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.293
p=0.0315
p=0.1441
p=0.5664
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAD1 | 235253_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=8e−04
p=0.446
p=0.7525
p=0.6795
SmallAirway2
LargeAirway2
Trachea
Macrophage2

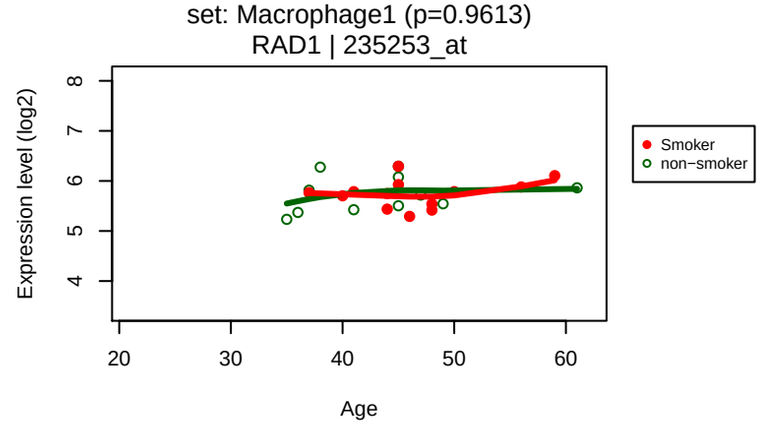

set: SmallAirway1 (p=0.0122)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker

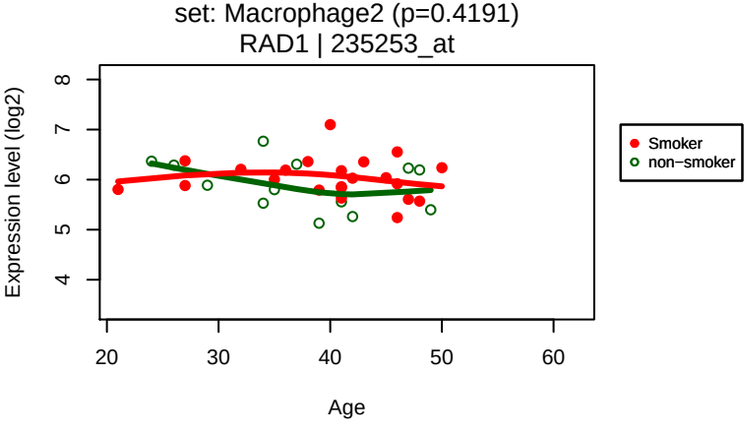

set: SmallAirway2 (p=0.022)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker

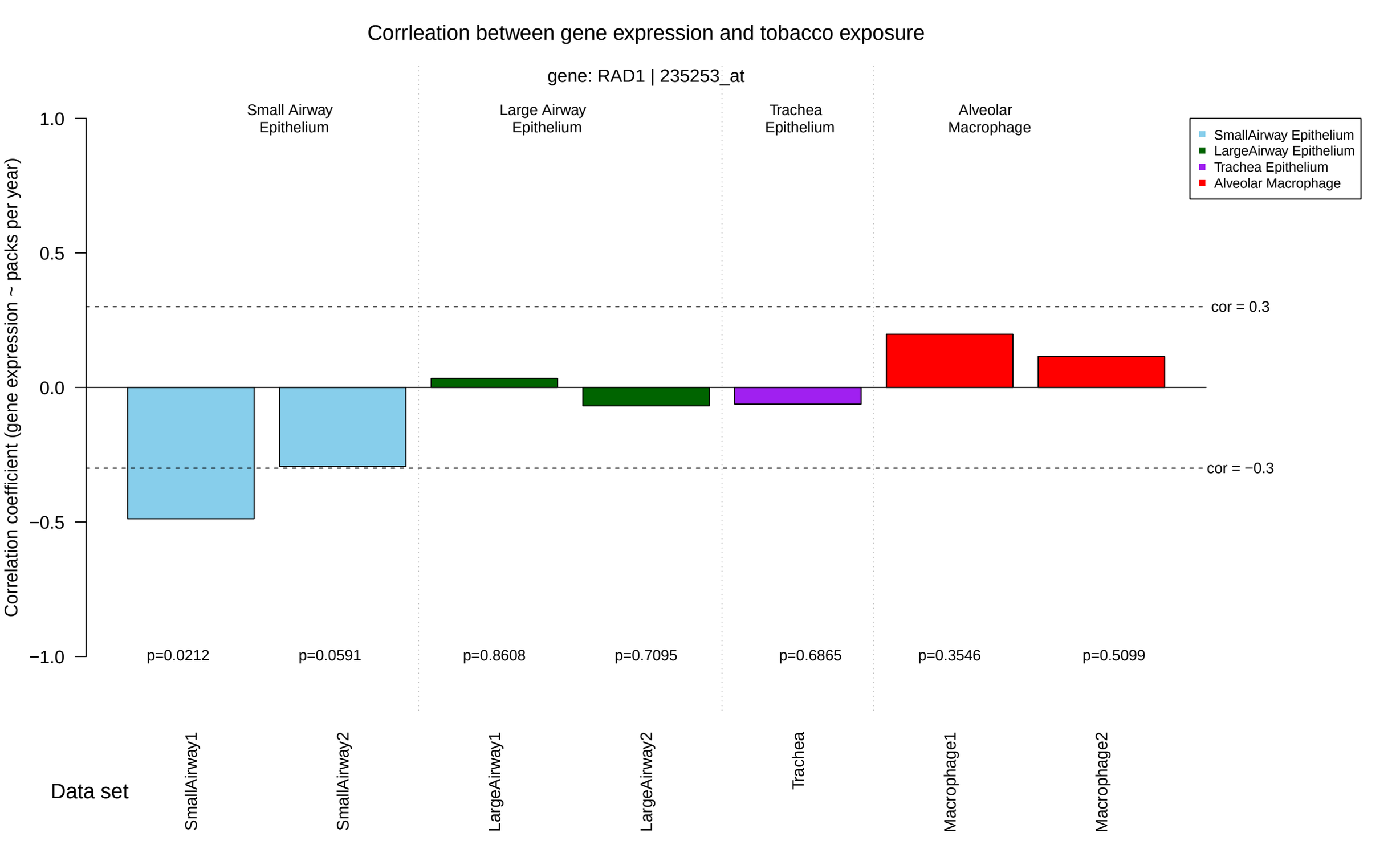

set: LargeAirway1 (p=0.2236)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1036)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4949)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9613)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4191)
RAD1 | 235253_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAD1 | 235253_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0212
p=0.0591
p=0.8608
p=0.7095
p=0.6865
p=0.3546
p=0.5099
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2